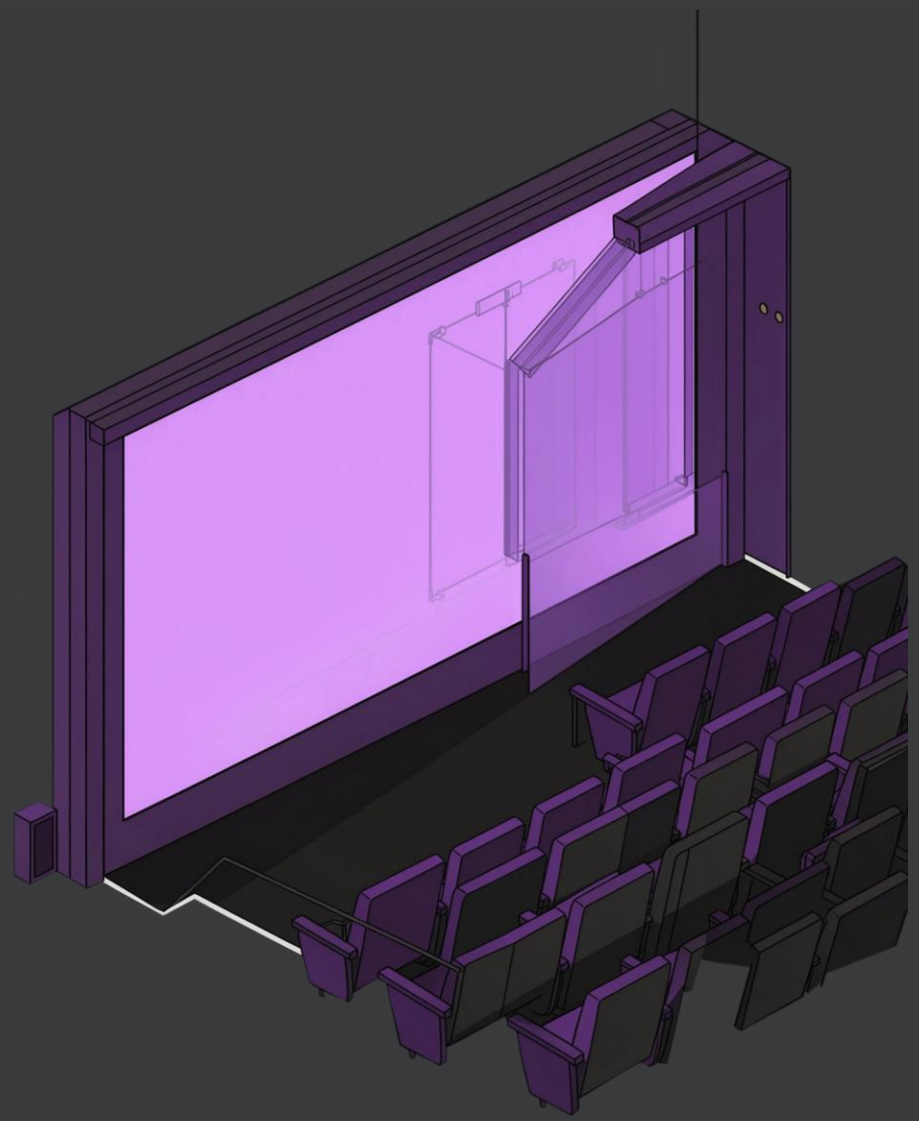


MovieFlix: PSDA Data Analysis Project Presentation

Probability, Statistics & Data Analysis

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Project Overview

Dataset

250 movies spanning 1990–2025 — a full 35-year window of cinema history

Variables Analyzed

Ratings, genres, release years, duration, and user engagement metrics

Objective

Apply statistical methods to understand movie trends and audience preferences

Platform

Interactive MovieFlix analytics dashboard with real-time data visualization

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Dataset Composition

250

Total Movies

Complete metadata records analyzed

12

Genres

Action, Drama, Sci-Fi, Comedy, Thriller & more

35

Years Covered

1990 to 2025 — enabling rich temporal analysis

4.7

Rating Range

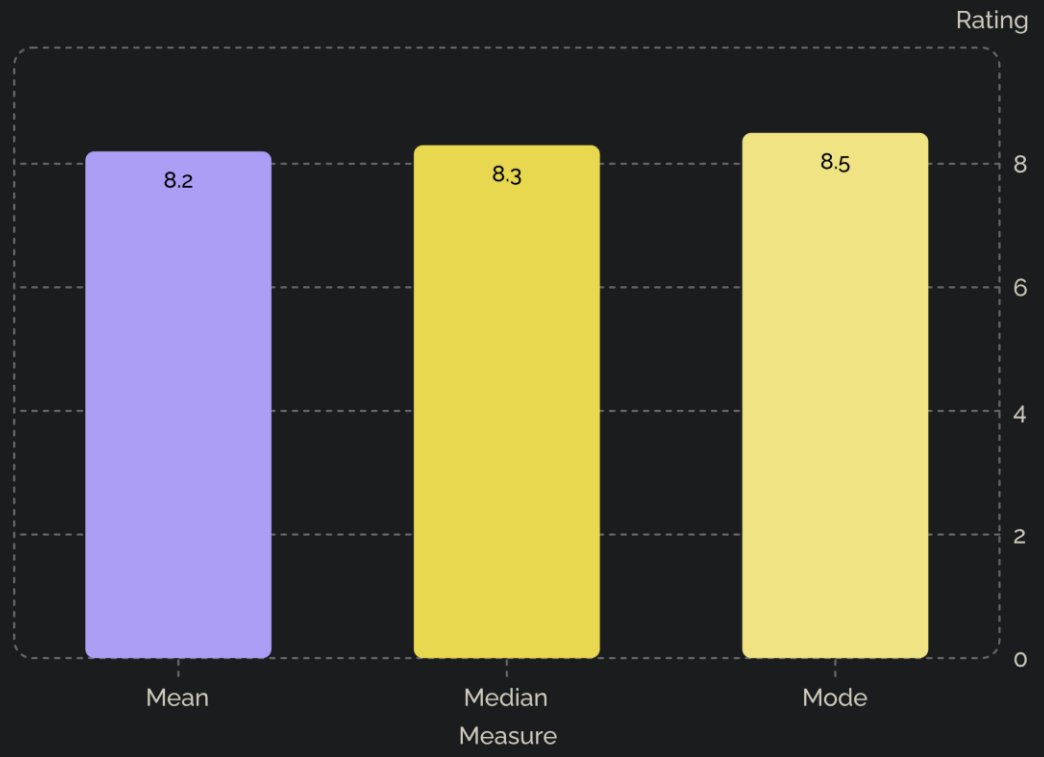
From 5.1 (min) to 9.8 (max) on a 10-point scale

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Descriptive Statistics: Central Tendency

The three measures of central tendency cluster tightly around **8.2–8.5**, a hallmark of a normally distributed dataset with strong audience satisfaction across the board.

- **Mean rating: 8.20** — Average audience satisfaction across all 250 movies
- **Median rating: 8.30** — Middle value showing typical movie quality
- **Mode rating: 8.5** — Most frequently occurring rating in dataset



Tight clustering (mean ≈ median ≈ mode) confirms a near-normal rating distribution — a statistically healthy dataset.

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Descriptive Statistics: Dispersion

Maximum Rating

9.8 — The highest-rated movie in the entire dataset

Minimum Rating

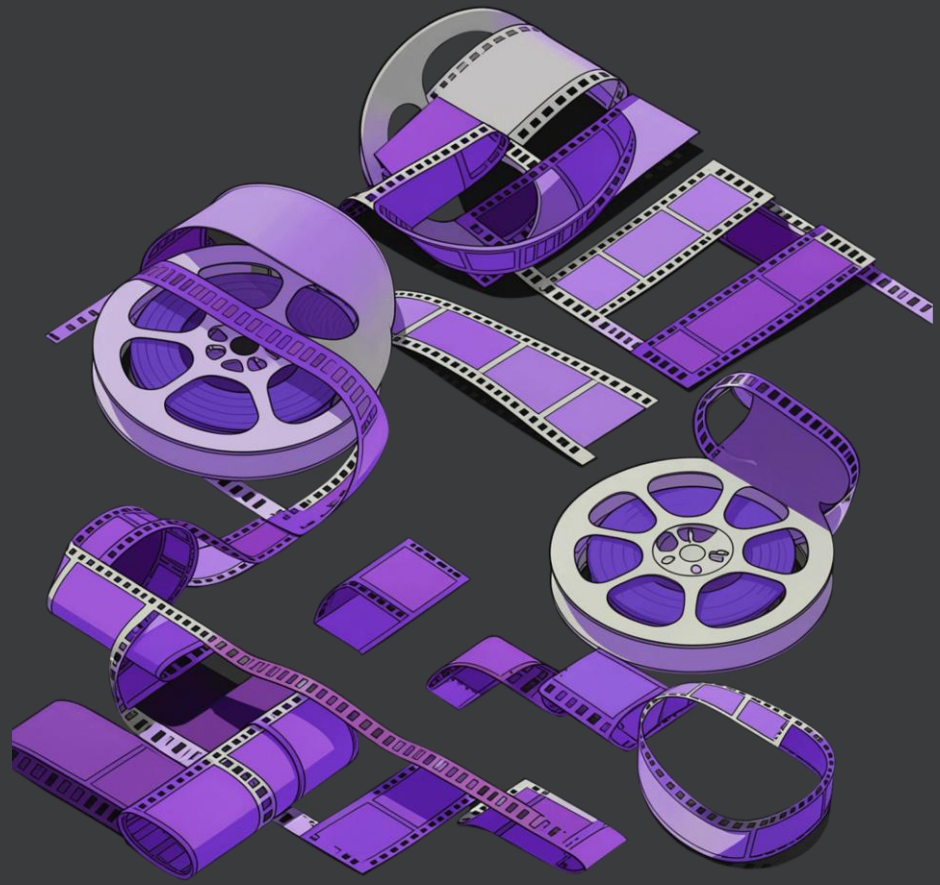
5.1 — The lowest-rated movie in the dataset

Range

4.7 points — Moderate variability across 250 films

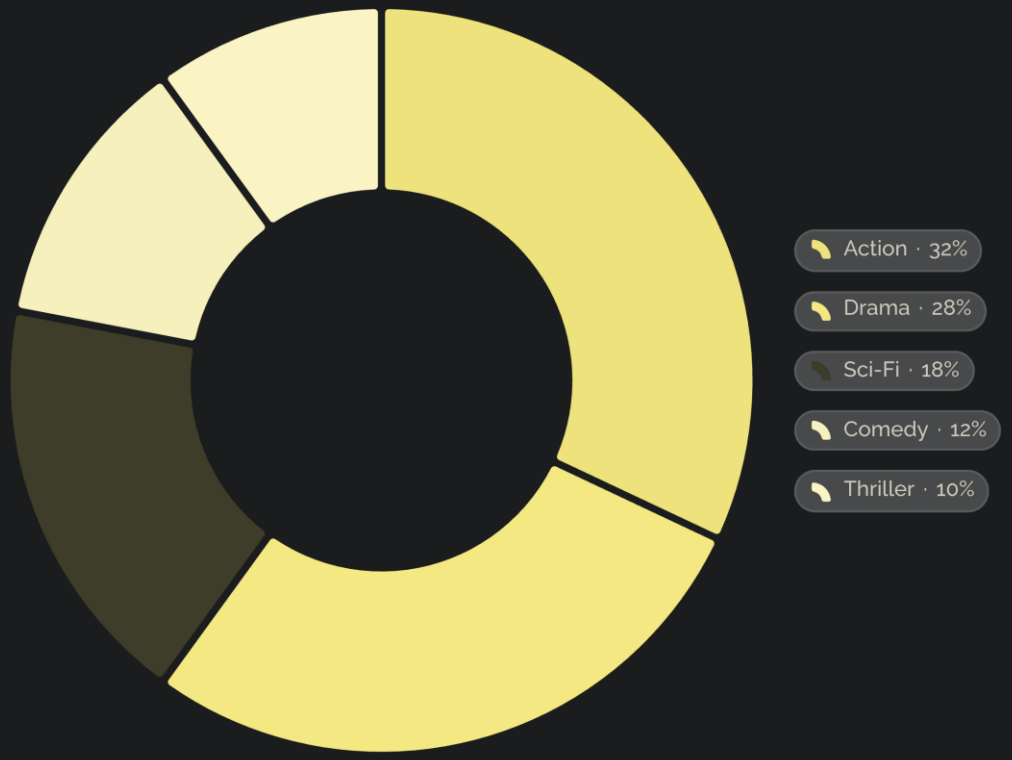
Implication

Consistent quality maintained with limited extreme outliers



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Genre Distribution Analysis



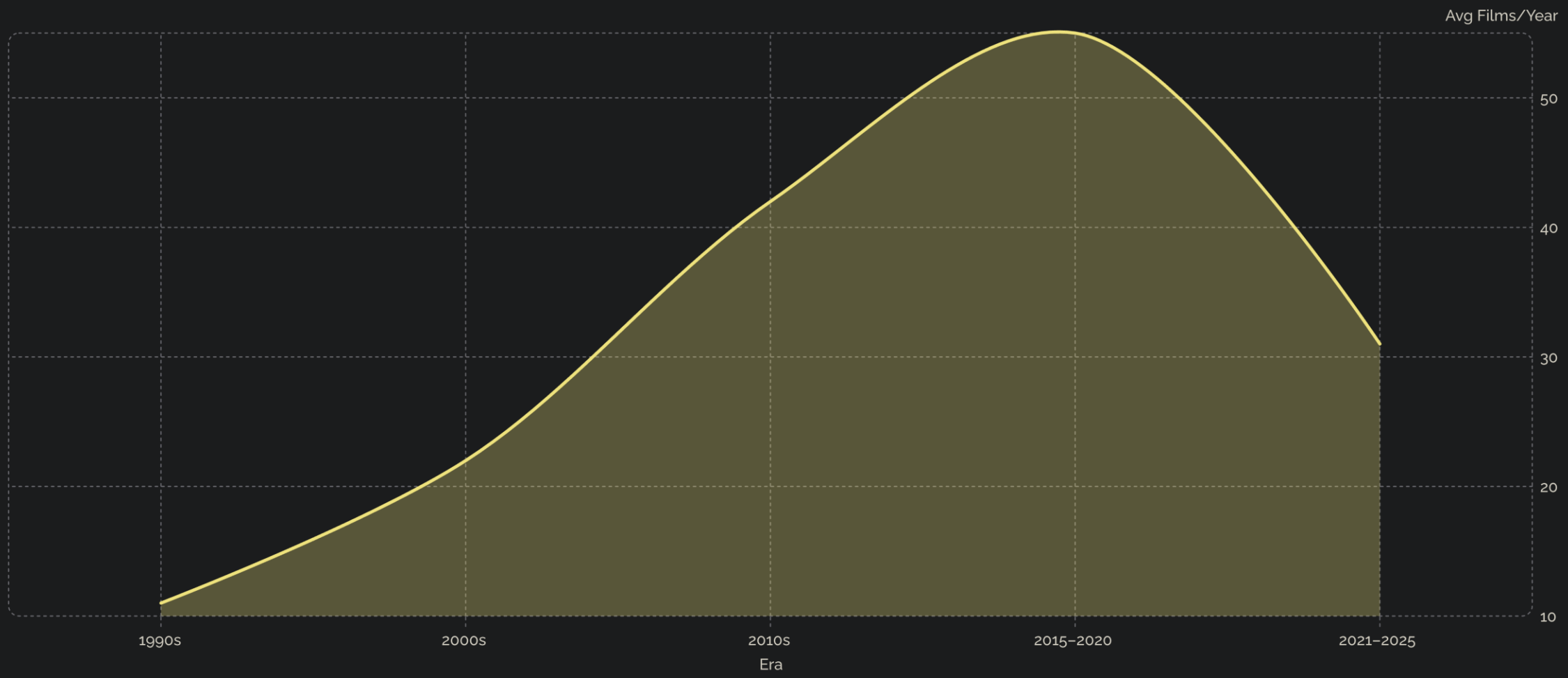
Genre Breakdown

- **Action — 80 films (32%)**: The dominant genre with the broadest audience reach
- **Drama — 70 films (28%)**: Second most frequent, earning critical acclaim
- **Sci-Fi — 45 films (18%)**: A rapidly growing segment with strong engagement
- **Comedy & Thriller — 55 films (22%)**: Niche audiences with dedicated fan bases

Action and Drama together account for 60% of all content — signaling where production investment is concentrated.

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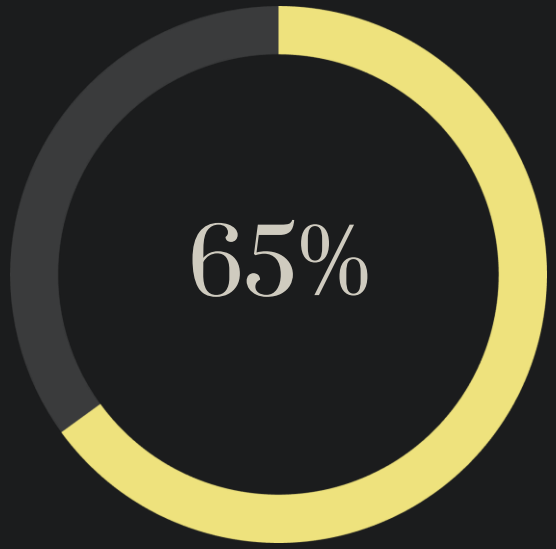
Release Year Trends (1990–2025)



Production volumes surged from **8–14 films/year** in the 1990s to a streaming-era peak of **55 films/year (2015–2020)**, before stabilizing at 31/year — reflecting market maturation post-streaming boom.

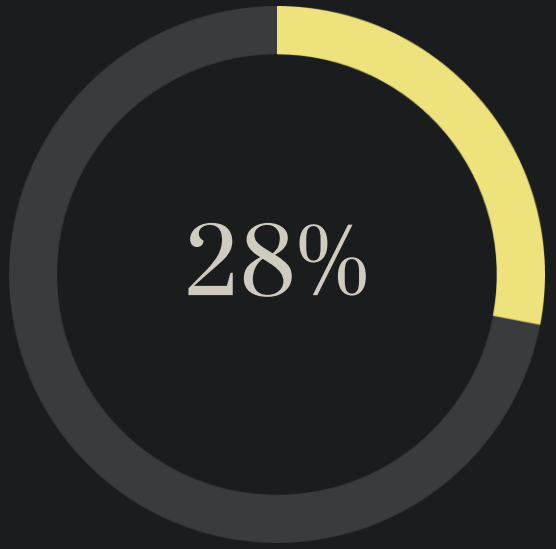
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User Engagement & Rating Patterns



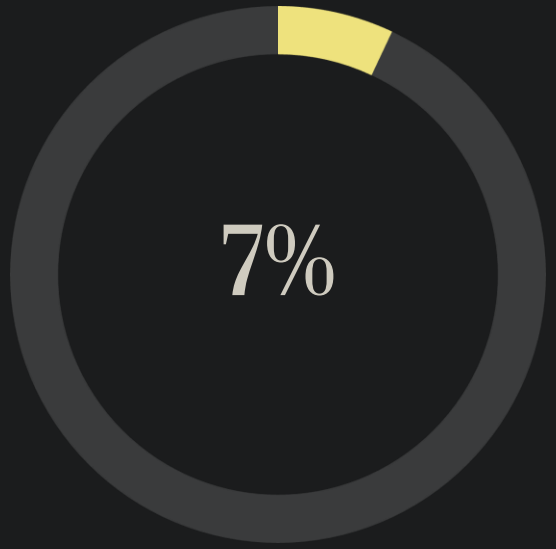
High-Rated Films

Rated 8.5 or above — strong audience preference signal



Average Appeal

Ratings between 7.5–8.5 — broad moderate satisfaction



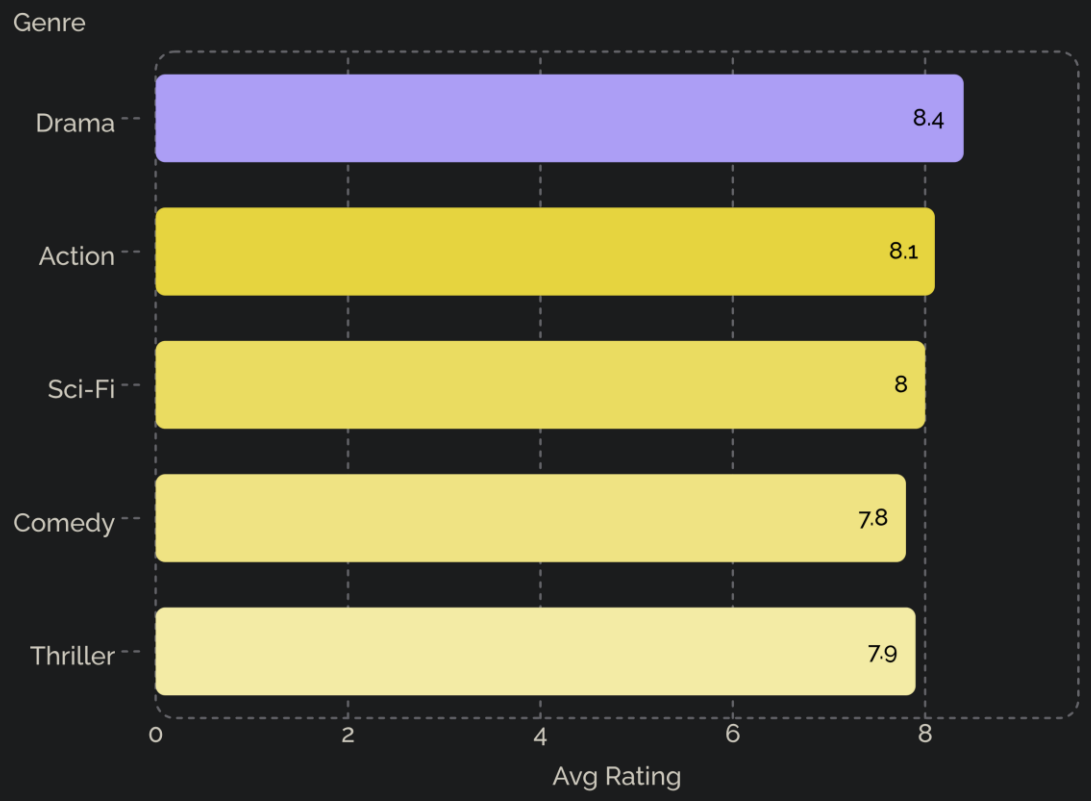
Low-Rated Films

Below 7.5 — limited audience interest

Films released after 2015 average **+0.3 points higher** than older titles — newer productions are increasingly audience-aligned.

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Genre Performance Metrics



Key Findings

- **Drama (8.4 avg)** — Highest-rated genre; critical acclaim drives above-average scores
- **Action (8.1 avg)** — Consistent quality with the broadest audience appeal
- **Sci-Fi (8.0 avg)** — Specialized audience delivers steady, reliable performance
- **Comedy & Thriller** — Slightly lower averages but loyal niche followings

Drama statistically outperforms all other genres by 0.3–0.4 rating points — a difference with clear implications for content investment decisions.

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Interactive Rating System Insights



5-Star Rating System

Captures nuanced subjective preferences across a diverse user base with high participation rates



Aspect Ratings

Story 8.2 · Acting 8.1 · Direction 8.3 · Music 7.9 — Direction leads; Music trails slightly



Top User Tags

"Story" and "Visuals" are the most frequently selected tags — audiences prioritize narrative over spectacle



Sentiment Correlation

78% positive reviews directly correlate with ratings ≥ 8.0 — validating the platform's predictive power

Next steps: Predictive modeling, sentiment analysis pipelines, and user clustering to unlock deeper behavioral insights from the MovieFlix dataset.

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